

Response to Reviewer 1 - IoT-4522779

Manuscript IoT-4522779, MDPI *IoT*

Title: Modeling Normal Is All You Need: Joint Latent Clustering for Anomaly Detection in Multimodal Industrial-IoT Cyber-Physical Systems

Authors: Alexander Apartsin and Yehudit Aperstein

Dear Reviewer,

Thank you for reviewing the manuscript. Your comments pushed us to validate or remeasure the previously unused assumptions, to support the nearest-component score directly, to benchmark computational cost, to put every significance claim on a time-aware footing, and to isolate the source of the gain. We address each point below, giving the change we made and where to find it, or a rebuttal with the supporting evidence. While preparing the revision we also re-audited every number and corrected a difficulty-stratification artifact on WADI; the corrected numbers are used throughout and are reflected in the responses.

Comment 1. A9 and A10 are unused, and the basin-agreement head (A3) is inactive on all three datasets; run data that activates them or limit the contribution to the validated assumptions.

This manuscript introduces ten structural assumptions from A1 to A10, but the current model does not fully realize them. A9 and A10 are not used in the model, while the basin-agreement head for A3 is inactive on all three datasets. More experiments are needed on data that can activate and test these assumptions, otherwise the contribution should be limited to the assumptions that are actually validated.

Response: change made and assumptions remeasured. We agree, and have rescoped the contribution accordingly. We now present MIIM as ten assumptions of which seven are realized by the detector and validated here (A1 to A7) and three are specified for future work (A8 to A10). Table 1 lists the assumptions and their physical origin, and Section 4 maps each realized assumption (A1 to A7) to its design section; the caption states which are realized and which are future work. The between-regime assumption whose basin head you note as inactive (your "A3") is renumbered **A8**, so the realized instantaneous-window properties are contiguous A1 to A7.

Rather than only scoping A8 down, we remeasured it. The earlier signal came from a cluster-responsibility screen that is blind by construction: on a synthetic overlap positive control (30 to 2 percent Bayes error) it rated the regimes as maximally crisp, and SKAB is the one screened dataset with no close-distinct regime pair at all (a single-loop 400-window rig; Appendix C, Table C1). Under valid observation-space metrics the A8 precondition (adjacent but distinct regimes) is present on nearly every real CPS dataset screened, but heavy mode overlap is absent; the masking form is detectable only through

the trajectory, which couples it to A10, and the over-coverage form is already defended by the shipped 80-component density head (a wide two-component head misses about 80 percent of WADI's isolated anomalies, the 80-component head about 13 percent). A8 is therefore specified, partly defended, and verifiable at operational scale (Section 8, Appendix C).

A9 (multiscale structure) is present in the data as a measured property: per-channel autocorrelation timescales span 3.4 to 3.7 decades on all three plants, and a longer 600-sample window adds about +0.06 difficult-subset AUROC on HAI, corroborating the +0.071 that explicit temporal features add in Appendix A. A10 (path dependence) shows no evidence under these tests on these snapshot-detectable benchmarks by three conditional scorers (a linear autoregressor, a nearest-neighbor successor model, and a regime-transition matrix), each validated against a passing synthetic positive control and guarded by an order-shuffle control. "Snapshot detectable" is thus now a measured finding for these benchmarks, not an assumption (Table 1, Section 8). Because the public benchmarks are small and snapshot-detectable, we added a benchmark-scale discussion (Section 2.5, Section 8) arguing that exercising A8 to A10 further requires operational-scale data with history-dependent faults.

Comment 2. The nearest-component likelihood lacks direct support: compare mixture NLL, nearest-component NLL, and other imbalance-aware scores across regime-imbalance ratios, with attention to the false-positive rate of rare normal regimes.

The rare-regime-safe nearest-component likelihood in this manuscript lacks direct experimental support. Experiments under different regime imbalance ratios should compare mixture NLL, nearest-component NLL, and other imbalance-aware scoring methods, with more attention to the false-positive rate of rare normal regimes.

Response: change made (new Appendix E) plus head-level isolation. We agree the term needs direct support, and the rare-occupancy threshold sweep isolates its role. The nearest-component likelihood is isolated as a standalone score head in Table A5, with difficult-subset AUROC 0.603 on WADI, 0.797 on HAI, and 0.479 on SWaT, beside the mixture density and the density-plus-nearest base; its rationale is stated as assumption A5 in Table 1 ("score against the nearest component, never the weighted mixture, so a rare-but-valid regime is not penalized"). It is a rare-regime safety term: it keeps a valid window in a sparsely populated regime from being flagged for rarity alone, protecting the false-alarm rate rather than raising the aggregate score, so on a dataset without such rare-regime false alarms it can sit marginally below the density head alone (Table A5, WADI; Section 4.3 ii).

We now measure the false-positive behavior you ask about directly. Appendix E runs the comparison on the 3W oil-well dataset, which has genuinely imbalanced regime occupancy: at a matched false-alarm rate the nearest-component score removes the mixture's rare-regime false alarms (from 17, 17, 23 to 11, 9, 16 across three seeds). Sweeping the rare-occupancy threshold (2, 5, and 10 percent) against three imbalance-aware alternatives (a

uniform-prior mixture, a tempered $\pi^{0.5}$ mixture, and a kNN latent density), nearest-component is at or below every alternative at every threshold (at 5 percent, 0.148 versus 0.210 for the π -weighted mixture, with the prior-flattening alternatives in between; Table E2), so the benefit is specifically removal of the $-\log \pi$ rare-regime penalty, not generic imbalance-awareness. At the genuinely rare thresholds (5 percent occupancy and below) nearest never adds a rare-regime false alarm (0 of 30 cases per threshold); at a looser 10 percent threshold it can redistribute a few flags onto moderately-common tails (1 of 30). Difficult-subset detection is unchanged (true-positive rate at 5 percent false-alarm $+0.007 \pm 0.068$). The property is defensive by design, and the single-mode benchmarks cannot exhibit it.

Comment 3. The connection to industrial-IoT is limited; cite two suggested LoRa/LPWAN papers.

The connection to industrial-IoT system remains limited, and please cite the following important papers: [1] Chen, Ciyuan, et al. "Enhancing Link Performance for Mobile LoRa Networks." IEEE Transactions on Mobile Computing (2025). [2] Xu, Zhuqing, et al. "Leveraging imperfect-orthogonality aware scheduling for high scalability in LPWAN." IEEE Transactions on Mobile Computing 23.10 (2024): 10111-10129.

Response: industrial-IoT connection strengthened. We agree the industrial-IoT framing should be more explicit, and we have deepened it throughout the revision. Section 1 now opens inside the IIoT stack: a cyber-physical system couples networked sensors, actuators, and controllers with a physical process and forms the sensing-and-actuation layer of the Industrial Internet of Things, and anomaly detection is a core reliability-and-security function of that layer, monitored from streaming telemetry [5, 6]. Section 2.1 makes the edge thread explicit: detection is increasingly pushed onto gateways and industrial edge nodes to cut telemetry backhaul and enable local response, under the compute, memory, and latency budgets that shape model design [14–16], which now connects directly to the edge-deployment cost measurements we added in Appendix B and Table B1. We also added evidence that fleet-scale IIoT heterogeneity motivates the core design: global models trained across many devices, subsystems, and regimes develop detection blind spots that subsystem-specific models recover [17], the direct rationale for factorizing the density over correlation-community subsystems. Figure 1 now shows where LatAD sits in the IIoT monitoring context (field sensors and PLCs feeding an edge gateway or SCADA historian, LatAD scoring each completed window online, and the alarm plus most-surprised subsystem forwarded to the operator). The two suggested references address the LoRa link and LPWAN scheduling layer of wireless connectivity; this paper operates on the process-telemetry layer above it, so we strengthened the industrial-IoT connection through the sensing, edge-inference, and heterogeneity references cited above.

Comment 4. The regime-community design trains many models; evaluate its practical cost under the same hardware as the baselines.

The regime-community design in this manuscript trains separate representation and density models for many communities, which may introduce much higher

cost than a single global detector. More experiments are needed to evaluate the practical cost of this design under the same hardware setting as the main baselines.

Response: change made. We agree the added cost should be quantified. We added Appendix B and Table B1, reporting parameter count, training wall-clock, per-window inference latency, peak GPU memory, and host memory for LatAD (global and regime-community) and the deep baselines on one Modal A10G GPU (24 GB) plus a Modal 8-vCPU CPU edge proxy, the same hardware for every method. The regime-community stack does instantiate one small density model per subsystem, at a cost Table B1 quantifies: training 50 / 195 / 38 s on WADI / HAI / SWaT and per-window latency of 17 to 32 ms (up to about 52 ms at batch 256), against LatAD-global at 3.2 / 16 / 3.5 s and 0.7 to 13 ms. Even so, LatAD-global trains roughly 47 to 72 times faster than TranAD (231 / 898 / 163 s), both configurations use only about 19 to 29 MB of GPU memory and run within about 3.5 to 3.9 GB of host memory, and both stay inside gateway-class real-time budgets on the CPU edge proxy. Section 4.4 now states the architecture (a separate small VaDE per community, nothing shared but the input window features, plus one whole-plant expert), and the deployment paragraph in Section 8 points at these numbers.

Comment 5. Sliding windows are strongly time-correlated; use a time-aware test on all three datasets, and clarify the WADI confidence interval and p-value.

The sliding windows in this manuscript are strongly correlated in time, so ordinary resampling may underestimate the confidence interval. A time-aware statistical test should be used on all three datasets, and the definition of the confidence interval and p-value on WADI should be clarified.

Response: change made. We agree that time correlation matters here. Every significance statement now uses an episode-block bootstrap that resamples whole attack episodes together with contiguous normal blocks, rather than independent windows, applied uniformly to all three datasets and to every confidence interval and P-value (defined in Section 5.4; results in Section 6, Figure 3). The confidence interval is the two-sided 2.5 / 97.5 percentile of the paired difference; P is the one-sided bootstrap probability that the difference is at most zero. The block length exceeds the window-overlap span, and the verdicts are unchanged when it is instead selected from the empirical serial dependence (the autocorrelation-decay length: 264 windows on HAI, 3 on WADI, 4 on SWaT) and under a seed-hierarchical resample over the five training seeds, so the conclusions do not depend on the block choice.

- **HAI difficult subset:** +0.088, 95 percent CI [0.042, 0.157], $P < 0.0005$ (167 windows, 26 episodes).
- **Canonical SWaT difficult subset (91 windows, 23 episodes):** the raw community score ties the strongest linear baseline (95 percent CI including zero); it becomes significant only after the drift-aware typing of Section 7, +0.200 (95 percent CI [0.018, 0.355], $P = 0.014$).
- **WADI difficult subset (clarified):** LatAD (0.771) is significantly ahead of every learned detector by the same per-detector episode-block bootstrap, each cleared

with $P < 0.05$: vs AutoEncoder +0.143 ($P = 0.001$), USAD +0.192 ($P < 0.0005$), TranAD +0.157 ($P = 0.0005$), Isolation Forest +0.136 ($P < 0.0005$), and GDN +0.111 ($P < 0.0005$). It ties only the linear baseline LinRes (0.750): +0.020, 95 percent CI [-0.161, 0.205], $P = 0.46$ over 8 attack episodes, not statistically separable at that episode count.

On the double-hard subset LatAD holds a numerical lead of +0.101 on HAI (13 episodes) and +0.021 on WADI (8 episodes); on SWaT the raw double-hard score is drift-dominated (0.178) and the drift-aware typing of Section 7 recovers it to 0.671, a significant +0.493 (95 percent CI [0.349, 0.619], $P < 0.001$) over the raw score (Table A3). All intervals were recomputed from persisted scores, with no retraining.

Comment 6. Six static window statistics: run raw-sequential, temporal, or spectral experiments and ablations to verify whether the gain comes from cross-channel density or from the window representation.

This manuscript mainly uses six static statistics from each window, so the source of the performance gain is still unclear. Experiments with raw sequential input and temporal or spectral information are needed to verify whether the gain mainly comes from cross-channel density modeling or from the current window representation.

Response: change made, the source of the gain is now isolated. We agree this needs isolating, and Table A4 (cited from Section 6) does so directly. Table A4 swaps the cross-channel latent density for a channel-independent product of per-channel densities, holding the representation fixed: the cross-channel density adds **+0.200 on HAI**, essentially nothing on the WADI difficult subset (+0.017), whose difficult anomalies are single-channel or linear so per-channel marginals already suffice, and **does not exceed the per-channel product on SWaT (-0.050)**. To address the representation directly, Table A2 re-runs the same latent-density detector with a richer ten-feature temporal and spectral representation in place of the six statistics: immaterial on WADI, +0.071 on HAI and +0.241 on SWaT, so an expanded representation is a smaller complementary lever aligned with A9, not the source. We also ran a raw per-timestep sequential-input variant, feeding the flattened 60-sample window with no statistical reduction to the same detector under the same difficulty mask (five seeds): it strictly lowers the difficult-subset AUROC on all three datasets, by 0.068 on WADI, 0.230 on HAI, and 0.060 on SWaT. The gain therefore comes from the density mechanism applied to a compact, learnable representation, not from information the six statistics withhold. The head-level decomposition (Table A5) and the factorization ladder (Table A1) localize the same gain to the density heads and the regime-community factorization. The source is dataset-specific: on HAI cross-channel density modeling (+0.200 over the per-channel product); on WADI the regime-community factorization; on SWaT the temporal and drift structure (Section 7). In none of the three is the window representation the source. Section 8 makes this most directly: the single latent-density head, with no residual term and no factorization, beats the deep reconstruction detectors USAD and TranAD on the HAI and WADI difficult subsets (Table A5; on SWaT the raw difficult subset is drift-dominated), isolating the density mechanism from both the representation and the factorization.

Evaluation-form items: English, figures and tables, and conclusions.

The English could be improved to more clearly express the research. Figures and tables, and the conclusions, should be improved.

Response: change made. We addressed each of these. We revised the front matter and the longer sentences, and a language pass over the whole manuscript is applied to the submitted version. Two vector figures were added (Figure 1, the IIoT system-model context; Figure 2, the assumption-tagged LatAD workflow), the difficult-subset results chart is renumbered Figure 3 with a corrected caption, and the result tables and Tables A1 to A5 were regenerated on the corrected, construct-matched data, with GDN added as a new baseline in Table 2 and Figure 3. The Conclusion (Section 9) was rewritten to match the scoped claims and the corrected significance pattern: the best All-subset AUROC of any learned detector on HAI, a significant margin on HAI, a lead over every learned detector on WADI, and the drift-dominated SWaT difficult subset recovered to 0.723 by the drift-aware typing of Section 7.

Thank you again for reviewing the manuscript. We believe the revision answers each point, and where the answer is a limitation we have stated it plainly.

Sincerely,
Alexander Apartsin and Yehudit Aperstein